

Connecting the Custom OCI Transport to Real Oracle OCI

This guide explains the **manual steps** to point PilotFish's custom `OciObjectStorageTransport` at a real **Oracle Cloud Infrastructure Object Storage** tenancy (not the local `floci-oci` emulator).

Module FQCN: `com.pilotfish.eip.modules.oci.OciObjectStorageTransport`. Demo route today uses `floci` endpoint `http://floci-oci:4599` — that override is what you remove/change for production Oracle.

Naming note: OCI = Oracle Cloud Infrastructure. This is not Amazon S3.

1. Prerequisites in Oracle Cloud Console

1.1 Tenancy & identity

- Access to an OCI tenancy (Console URL typically `https://cloud.oracle.com`).
- A compartment where Object Storage buckets are allowed (often root or a dedicated Integration compartment).
- An IAM **user** (or group) that PilotFish will authenticate as, with Object Storage write permissions.

1.2 Create / confirm the bucket

- Console → **Storage** → **Buckets** → **Create Bucket** (or reuse an existing bucket).
- Note the bucket **name** exactly (case-sensitive).
- Note the **Object Storage namespace** (shown on the Bucket details page / Tenancy details). CLI: `oci os ns get`.
- Decide object naming (demo uses `835_{ST}_{timestamp}.json`).

1.3 IAM policy (minimum for PutObject)

Attach a policy to the group that contains the API-key user. Example (adjust compartment name):

```
Allow group PilotFishIntegrators to manage objects in compartment Integration
Allow group PilotFishIntegrators to read buckets in compartment Integration
```

For least privilege, prefer `manage objects` (or `OBJECT_CREATE` / `OBJECT_OVERWRITE` style statements) scoped to one bucket if your org uses finer IAM.

2. Create an API signing key (Console)

- Console → **Identity & Security** → **Users** → select the integration user.
 - **API Keys** → **Add API Key**.
 - Choose **Generate API Key Pair** (easiest) or upload a public key you generated with OpenSSL.
 - Download / save the **private key PEM** immediately (Oracle only stores the public key).
 - Copy the **Configuration File Preview** values: user OCID, fingerprint, tenancy OCID, region.
- Store the PEM on the eiPlatform host in a protected path, e.g. `/opt/pilotfish/secrets/oci_api_key.pem` (readable by the Tomcat/eiPlatform OS user only; mode 600).

3. Values to collect (checklist)

Item	Where to find it	Transport field
Region	Console region selector / config preview (e.g. us-ashburn-1)	OciRegion
Namespace	Tenancy / Bucket details; oci os ns get	OciNamespace
Bucket name	Storage → Buckets	OciBucket
Tenancy OCID	Profile → Tenancy → OCID / config preview	OciTenancyOcid
User OCID	User details / config preview	OciUserOcid
Fingerprint	API Keys list / config preview	OciFingerprint
Private key PEM path	File you downloaded/generated on the PF host	OciPrivateKeyPath
Service Endpoint Override	Leave blank for real OCI (demo only used floci URL)	OciEndpoint = empty

4. Manual PilotFish / eiPlatform changes

4.1 Confirm the custom module JAR is installed

- Ensure modules-oci-object-storage.jar is in WEB-INF/lib (or your licensed modules drop folder).
- Restart eiPlatform / Tomcat after placing the JAR.
- No floci container, throwaway key, or local emulator is required for real OCI.

4.2 Edit the route Transport (eiConsole or route.xml)

On route 2 – **Split ST JSON And OCI**, open the **OCI Object Storage PutObject** transport and set:

```
OciRegion = us-ashburn-1 # your region
OciNamespace = <your-namespace>
OciBucket = <your-bucket>
OciObjectName = {ognl:getAttribute('OciObjectName')} # keep as-is if demo processors set it
OciContentType = application/json
OciEndpoint = # CLEAR / empty for real Oracle
OciAuthMode = API Key
OciTenancyOcid = ocid1.tenancy.oc1..aaaa...
OciUserOcid = ocid1.user.oc1..aaaa...
OciFingerprint = aa:bb:cc:...
OciPrivateKeyPath = /opt/pilotfish/secrets/oci_api_key.pem
OciPrivateKeyPem = # leave empty when using file path
```

Also update any “display only” processors that still embed the floci URL (e.g. Set OCI Object URL attribute) so ops dashboards don't show the emulator host.

4.3 Networking / egress

- eiPlatform host must reach Oracle Object Storage HTTPS endpoints for the region (typically https://objectstorage.<region>.oraclecloud.com).
- Allow outbound 443 through corporate proxies/firewalls; if an HTTPS proxy is required, JVM proxy system properties (or a future transport proxy setting) must be configured.
- TLS trust store must trust public CA certs used by Oracle (default JVM cacerts is usually fine).

5. Validate outside PilotFish first (recommended)

Before changing the route, prove credentials with OCI CLI or a tiny SDK script on the same host:

```
# ~/.oci/config pointing at the same user/key/region
oci os ns get
oci os object put -ns <namespace> -bn <bucket> --file ./test.json --name smoke-test.json
```

If CLI PutObject fails (401/404/403), fix IAM/credentials/namespace/bucket before debugging PilotFish.

6. PilotFish smoke checklist

- Drop a multi-ST 835 through the existing SFTP → split → JSON pipeline.
- Confirm transport stage OCI Object Storage PutObject completes without retries.
- In Console → Bucket → Objects, verify one object per ST (or expected object names).
- Optional: inspect TX attributes set by the module: `com.pilotfish.oci.requestId`, `com.pilotfish.oci.eTag`, `com.pilotfish.oci.objectName`.

7. Demo → real OCI: what to stop using

Demo piece	Action for real OCI
floci-oci Docker service (:4599)	Stop / remove from compose; not used
oci-config/floci_key.pem throwaway key	Replace with real API private key; do not reuse floci OCIDs
OciEndpoint=http://floci-oci:4599	Clear this field so the SDK uses regional Oracle endpoints
floci-init bucket bootstrap	Create bucket/policies in Oracle Console (or Terraform) instead

8. Security recommendations

- Never commit private keys to git; mount secrets via volume / secret store.
- Prefer file path over `OciPrivateKeyPem` inline in `route.xml`.
- Rotate API keys periodically; revoke old fingerprints in Console.
- Scope IAM to one compartment/bucket; avoid tenancy-wide `manage all-resources`.
- This demo module currently supports **API Key** auth only — instance/resource principals would be a follow-on enhancement for OKE/Compute.

9. Common failures

Symptom	Likely cause
401 / NotAuthenticated	Wrong fingerprint, wrong private key, clock skew, or user OCID mismatch
404 BucketNotFound / NamespaceNotFound	Wrong namespace or bucket name/region
403 NotAuthorizedOrNotFound	IAM policy missing object write on that compartment/bucket
Connection timeout	Egress/firewall/proxy blocking regional Object Storage HTTPS
Still writing to floci	<code>OciEndpoint</code> still set to emulator URL

Bottom line

The custom module already speaks real OCI PutObject via the official Java SDK. To go from the local emulator to production Oracle: create bucket + IAM + API key in Console, place the PEM on the eiPlatform host, clear `OciEndpoint`, and paste the real region/namespace/bucket/tenancy/user/fingerprint into the transport. No Amazon S3 settings are involved.